

Cloud Security Threats and Solutions

1. Misconfigurations

- **Threat:** Misconfigurations in cloud resources (e.g., open storage buckets) can expose sensitive data.
 - **Example: Capital One (2019)** – Misconfigured AWS S3 bucket led to the exposure of 100M+ customer records.
 - **Solution:**
 - Perform regular security audits.
 - Use **Infrastructure as Code (IaC)**.
 - Enable **MFA** and **role-based access controls**.
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2. Data Leaks

- **Threat:** Improper permissions or insecure storage can result in data leaks.
 - **Example: Microsoft (2022)** – Misconfigured Azure Blob Storage exposed 2.4TB of internal data.
 - **Solution:**
 - Encrypt data **at rest and in transit**.
 - Implement **Zero Trust** security models.
 - Restrict **public access** to storage.
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3. Insecure APIs

- **Threat:** Insecure APIs can expose cloud data or allow unauthorized access.
- **Example: Facebook (2019)** – Poorly managed APIs exposed over 540M records.
- **Solution:**
 - Use **OAuth 2.0 & API gateways** for secure authentication.
 - Implement **rate limiting and logging**.
 - Regularly test APIs for vulnerabilities.